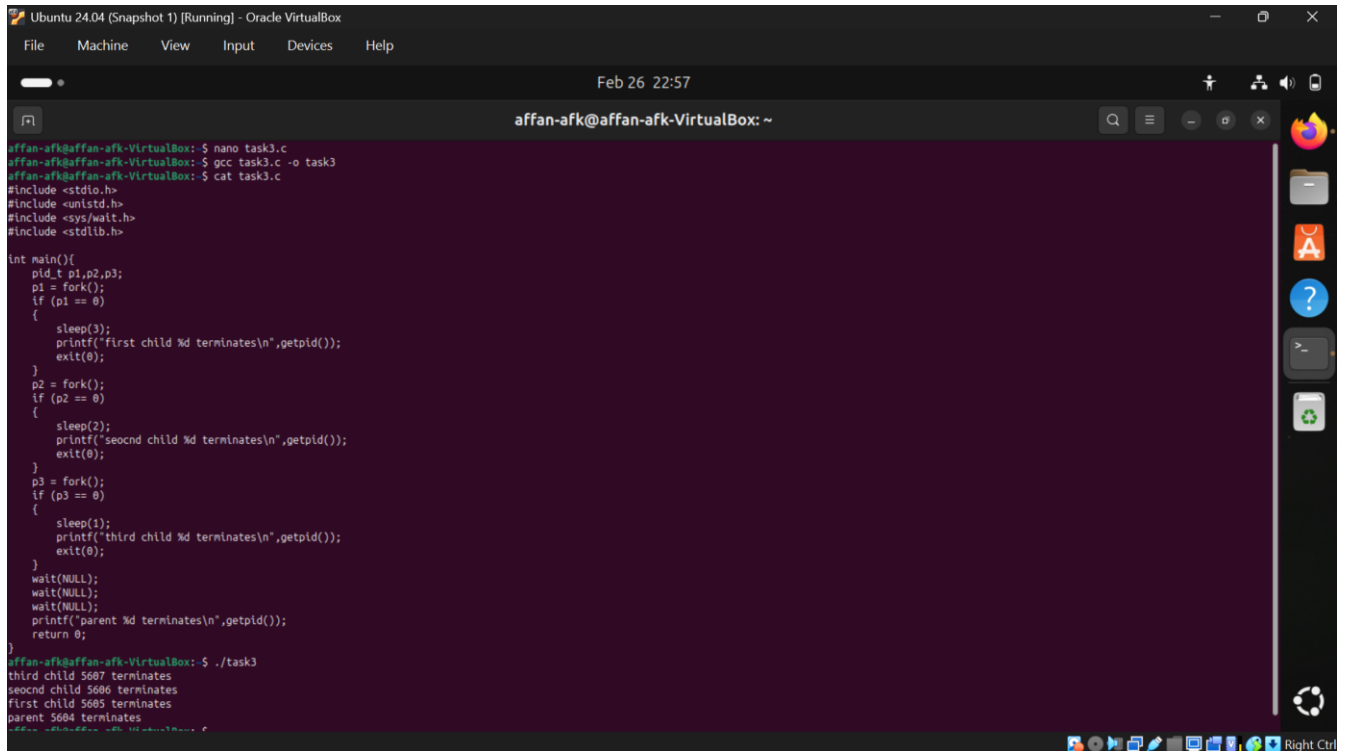


Task 3:



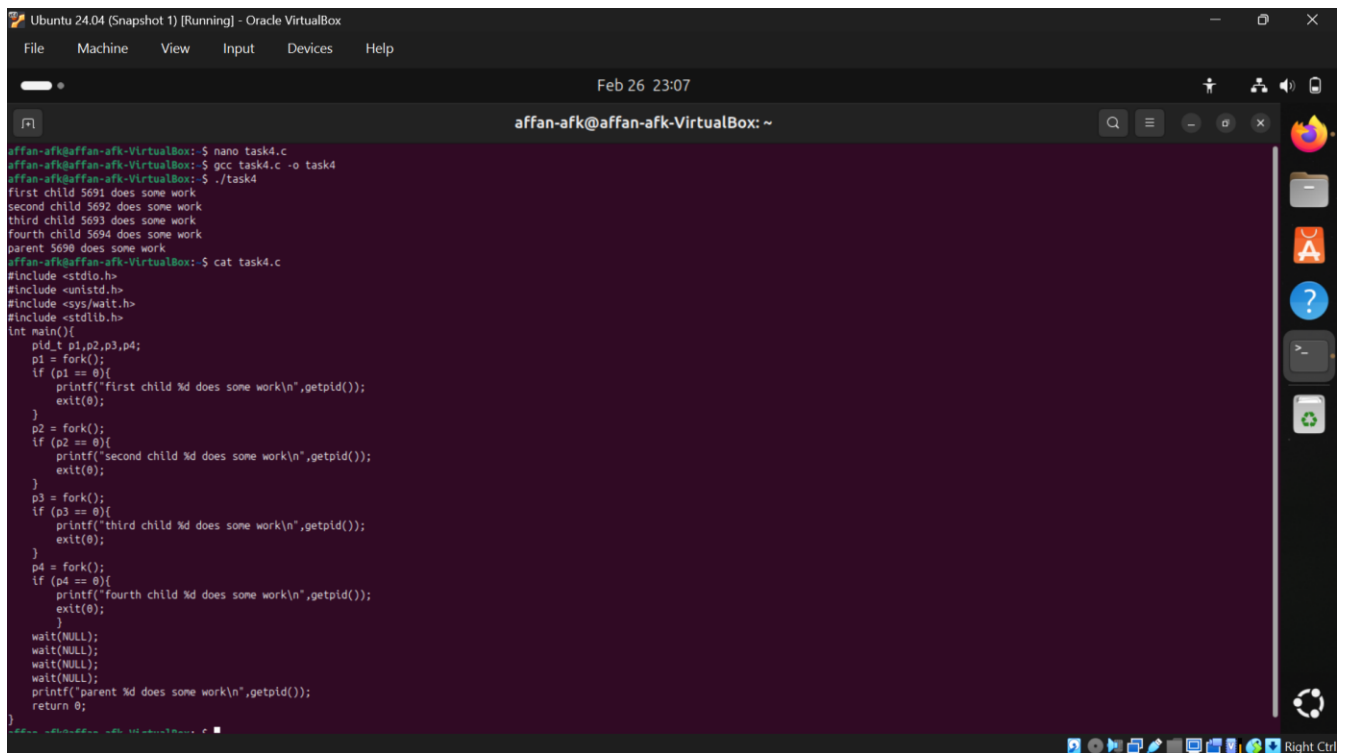
```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Feb 26 22:57

affan-afk@affan-afk-VirtualBox: ~
affan-afk@affan-afk-VirtualBox: $ nano task3.c
affan-afk@affan-afk-VirtualBox: $ gcc task3.c -o task3
affan-afk@affan-afk-VirtualBox: $ cat task3.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main(){
    pid_t p1,p2,p3;
    p1 = fork();
    if (p1 == 0)
    {
        sleep(3);
        printf("first child %d terminates\n",getpid());
        exit(0);
    }
    p2 = fork();
    if (p2 == 0)
    {
        sleep(2);
        printf("second child %d terminates\n",getpid());
        exit(0);
    }
    p3 = fork();
    if (p3 == 0)
    {
        sleep(1);
        printf("third child %d terminates\n",getpid());
        exit(0);
    }
    wait(NULL);
    wait(NULL);
    wait(NULL);
    printf("parent %d terminates\n",getpid());
    return 0;
}
affan-afk@affan-afk-VirtualBox: $ ./task3
third child 5687 terminates
second child 5686 terminates
first child 5685 terminates
parent 5684 terminates
```

Task 4:



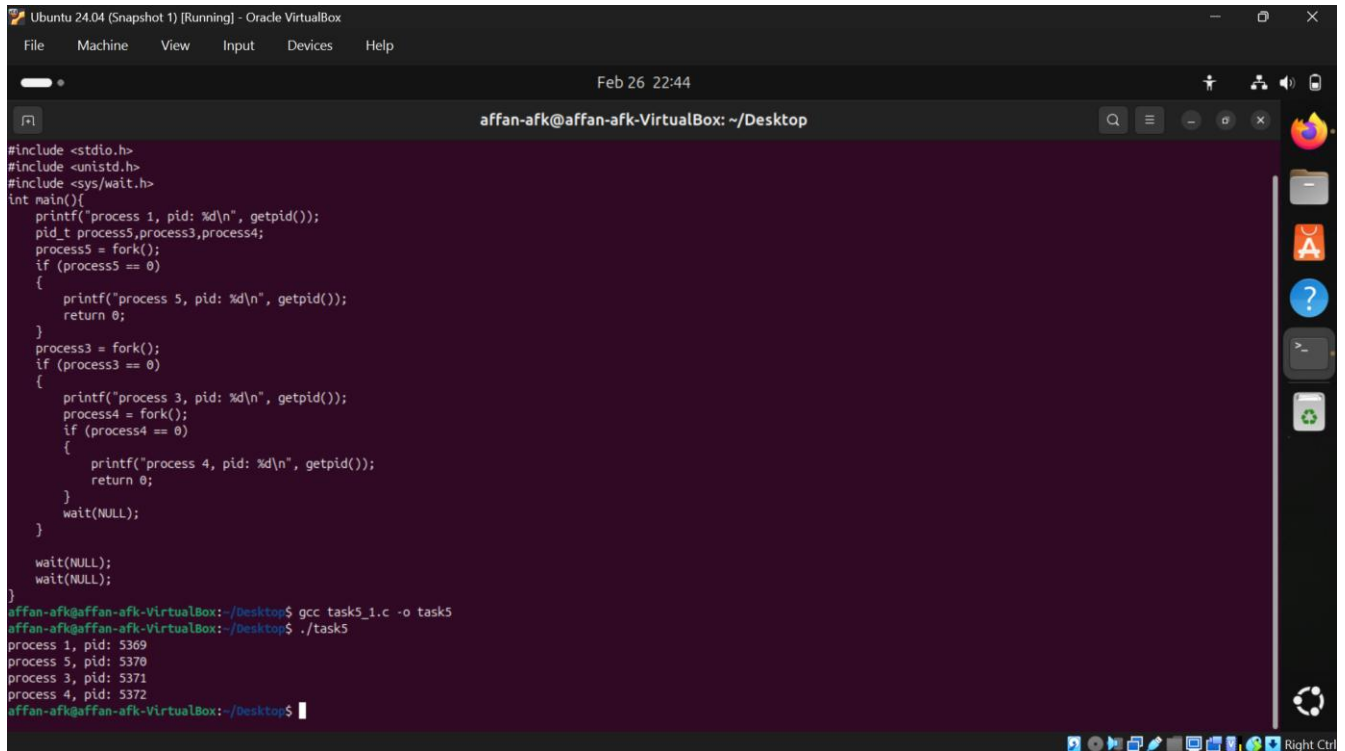
```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Feb 26 23:07

affan-afk@affan-afk-VirtualBox: ~
affan-afk@affan-afk-VirtualBox: $ nano task4.c
affan-afk@affan-afk-VirtualBox: $ gcc task4.c -o task4
affan-afk@affan-afk-VirtualBox: $ ./task4
first child 5691 does some work
second child 5692 does some work
third child 5693 does some work
fourth child 5694 does some work
parent 5690 does some work
affan-afk@affan-afk-VirtualBox: $ cat task4.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main(){
    pid_t p1,p2,p3,p4;
    p1 = fork();
    if (p1 == 0){
        printf("first child %d does some work\n",getpid());
        exit(0);
    }
    p2 = fork();
    if (p2 == 0){
        printf("second child %d does some work\n",getpid());
        exit(0);
    }
    p3 = fork();
    if (p3 == 0){
        printf("third child %d does some work\n",getpid());
        exit(0);
    }
    p4 = fork();
    if (p4 == 0){
        printf("fourth child %d does some work\n",getpid());
        exit(0);
    }
    wait(NULL);
    wait(NULL);
    wait(NULL);
    wait(NULL);
    printf("parent %d does some work\n",getpid());
    return 0;
}
```

Task 5 (1):



```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

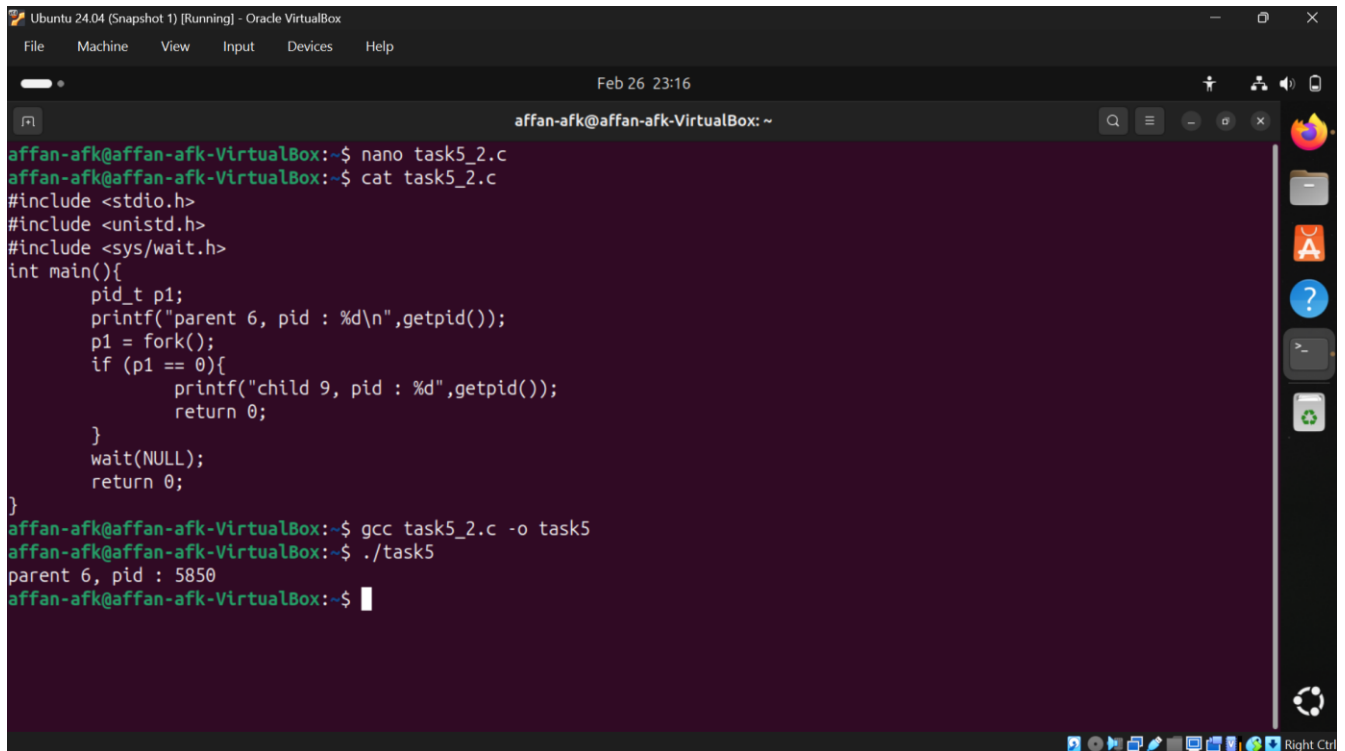
Feb 26 22:44

affan-afk@affan-afk-VirtualBox: ~/Desktop

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main(){
    printf("process 1, pid: %d\n", getpid());
    pid_t process5, process3, process4;
    process5 = fork();
    if (process5 == 0)
    {
        printf("process 5, pid: %d\n", getpid());
        return 0;
    }
    process3 = fork();
    if (process3 == 0)
    {
        printf("process 3, pid: %d\n", getpid());
        process4 = fork();
        if (process4 == 0)
        {
            printf("process 4, pid: %d\n", getpid());
            return 0;
        }
        wait(NULL);
    }
    wait(NULL);
    wait(NULL);
}

affan-afk@affan-afk-VirtualBox:~/Desktop$ gcc task5_1.c -o task5
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./task5
process 1, pid: 5369
process 5, pid: 5370
process 3, pid: 5371
process 4, pid: 5372
affan-afk@affan-afk-VirtualBox:~/Desktop$
```

Task 5_2:



```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Feb 26 23:16

affan-afk@affan-afk-VirtualBox: ~

affan-afk@affan-afk-VirtualBox:~$ nano task5_2.c
affan-afk@affan-afk-VirtualBox:~$ cat task5_2.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main(){
    pid_t p1;
    printf("parent 6, pid : %d\n",getpid());
    p1 = fork();
    if (p1 == 0){
        printf("child 9, pid : %d",getpid());
        return 0;
    }
    wait(NULL);
    return 0;
}

affan-afk@affan-afk-VirtualBox:~$ gcc task5_2.c -o task5
affan-afk@affan-afk-VirtualBox:~$ ./task5
parent 6, pid : 5850
affan-afk@affan-afk-VirtualBox:~$
```

3:

```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Feb 26 23:27

affan-afk@affan-afk-VirtualBox: ~/Desktop

affan-afk@affan-afk-VirtualBox:~/Desktop$ nano task5_3.c
affan-afk@affan-afk-VirtualBox:~/Desktop$ gcc task5_3.c -o task5
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat task5_3.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>
int main()
{
    pid_t p0,p7,p2;
    printf("in process 8 and my pid is %d\n",getpid());
    p0 = fork();
    if (p0 == 0)
    {
        printf("in process 0 and my pid is %d\n",getpid());
        p7 = fork();
        if (p7 == 0)
        {
            printf("in process 7 and my pid is %d\n",getpid());
            return 0;
        }
        p2 = fork();
        if (p2 == 0)
        {
            printf("in process 2 and my pid is %d\n",getpid());
            return 0;
        }
        wait(NULL);
        wait(NULL);
        return 0;
    }

    wait(NULL);
    return 0;
}
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./task5
in process 8 and my pid is 4356
in process 0 and my pid is 4357
in process 7 and my pid is 4358
in process 2 and my pid is 4359
affan-afk@affan-afk-VirtualBox:~/Desktop$
```

4:

```
Ubuntu 24.04 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Feb 26 23:43

affan-afk@affan-afk-VirtualBox: ~/Desktop

affan-afk@affan-afk-VirtualBox:~/Desktop$ nano task5_4.c
affan-afk@affan-afk-VirtualBox:~/Desktop$ gcc task5_4.c -o task5
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat task5_4.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>
int main()
{
    pid_t leaf1,inner_node,leaf2,leaf3;
    printf("in root and my pid is %d\n",getpid());
    inner_node = fork();
    if (inner_node == 0)
    {
        printf("in inner node and my pid is %d\n",getpid());
        leaf1 = fork();
        if (leaf1 == 0)
        {
            printf("in leaf 1 of inner node and my pid is %d\n",getpid());
            return 0;
        }
        leaf2 = fork();
        if (leaf2 == 0)
        {
            printf("in leaf 2 of inner node and my pid is %d\n",getpid());
            return 0;
        }
        wait(NULL);
        wait(NULL);
        return 0;
    }
    leaf3 = fork();
    if (leaf3 == 0)
    {
        printf("in leaf of root and my pid is %d\n",getpid());
        return 0;
    }
    wait(NULL);
    wait(NULL);
    return 0;
}
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./task5
in root and my pid is 4602
in leaf of root and my pid is 4604
in inner node and my pid is 4603
in leaf 1 of inner node and my pid is 4605
in leaf 2 of inner node and my pid is 4606
affan-afk@affan-afk-VirtualBox:~/Desktop$
```